

Introduction

- This report describes the design of a Natural Language Processing (NLP) system to analyse feedback from customers on products bought online. This report focused on Scenario 4 to understand the opinions of customers from large quantities of unstructured reviews. This scenario presents an easy way to use NLP techniques for extracting valuable insights into customer data for decision making.

PART 1 — Executive Summary

- The focus of this project was to develop an analysis system which would allow customers to provide input about products purchased through various retail websites utilizing natural language processing (NLP) technologies. E-commerce companies are increasingly facing the problem that customer comments/feedback provided via their websites are typically unstructured. Therefore, many of these firms lack tools to effectively analyze such feedback in order to better determine customer perceptions and product strengths or weaknesses. NLP technologies can convert large amounts of unstructured data into useful and structured formats (Jurafsky and Martin, 2023).
- A fully functional NLP analysis pipeline will be developed using the above approach. All customer review comments extracted from the Walmart Product Reviews dataset will undergo several phases of the pipeline. These phases include: text pre-processing; feature vectorization; information extraction; and language modeling. The system will also utilize techniques such as TF-IDF (term frequency-inverse document frequency) (Manning et al., 2008; Mikolov et al., 2013).
- Word2Vec to create an understanding of both how important certain words are within each review comment, and what semantic connections exist between words. Furthermore, named entity recognition (NER) will be used to pull out product related entities mentioned throughout all the customer reviews. Finally, n-grams (sequences of n items) will be created from customer comments to analyze patterns in the data. Representative text samples will then be generated based upon those patterns.

PART 2 — Text Pre-processing Strategy

- Text Preprocessing is a major component of the NLP pipeline for transforming an unstructured, raw set of customer reviews into a formatted structure that can be used to analyze the text. The Customer Product Feedback System will also need to process noisy data with features of informal language, misspellings, and extraneous punctuation and symbols associated with user generated content (Jurafsky and Martin, 2023).
- The first step in the process uses a combination of Regular Expressions (regex) for text cleaning. Cleaning involves the removal of URL's, special characters, extra punctuation, and non-alpha numeric symbols that provide no semantic value. For instance, many times in customer review data you will see multiple instances of a character (such as "goooooood") or excessive use of symbols (!!!!), and without normalizing these values could skew your analysis.
- After cleaning, the next stage of processing called Tokenization splits the text into individual word(s) or tokens. Tokenization creates an independent way to analyze each word, which provides a foundation for additional linguistic analysis. Tokenization also plays a major role in determining what commonly occurring terms exist in reviews such as quality, price, delivery etc... which are directly related to a products overall rating (Bird, Klein and Loper, 2009).
- When considering how to preprocess, one major decision factor is whether to utilize stemming versus lemmatization. Stemming takes the words down to their roots by removing suffixes (ex: running -> run) however it has been proven that stemming produces non linguistic results. On the other hand Lemmatization produces words back to their original dictionary form while preserving the same amount of meaning (ex: better -> good). For this project, since we want to preserve the semantics from our customers reviews and extract relevant information from them; I am going to choose lemmatization over stemming. (Manning et al., 2008).
- Removing Stop Words is another significant component of the preprocessing. Stop words are typically words that have very little significance within the context of the review (ex: and, is, the). Removing

these types of words minimizes unnecessary complexity and enhances computational speed. Nevertheless, there is a necessity for caution when eliminating stop words. Negating stop words (ex: not) greatly influence the positive/negative nature of a statement (ex: not good vs. good).

- Last but not least Text Normalization is performed. As part of Text Normalization, all text is converted to lower case and all words are standardized. Standardizing all words insures consistency through out the entire dataset and eliminates duplicate features caused by differing cases.

PART 3 — Vectorization & Representation

- A crucial part of NLP is converting text to numeric representations so that NLP can work with ML models. In Customer Product Feedback Systems, the ability to convert text to numbers effectively allows you to find key terms, recognize sentiment patterns, and understand the semantics of how customers talk about your products (Jurafsky and Martin, 2023).
- An easy way to do this is to use the Bag-of-Words (BOW) Model, where text is represented using only the number of times each word appears in a document. BOW is an easy way to convert text to numbers and also inexpensive from a computational perspective; however, it does not take into consideration the total frequency of the terms across all documents. The term "product" or "item", for example, might occur many times across all documents and while they tell us something about what was being reviewed, they tell us very little about what customers thought about their experiences with our products.
- The Term Frequency-Inverse Document Frequency (TF-IDF), addresses this problem. TF-IDF weights terms more heavily when they are infrequent across all documents but frequent in one particular document. Since TF-IDF weighs terms according to their significance to the review document, it will highlight important terms like "excellent," "broken," or "delayed" that indicate much more about a customer's experience with your product Manning, Raghavan and Schütze, 2008).

- While both BOW and TF-IDF count frequencies of words to represent text as numbers, Word2Vec goes further by representing each word as a dense vector in a continuous vector space. By doing this, Word2Vec identifies semantic relationships between words. For example, Word2Vec knows that "cheap" is semantically similar to "affordable," and that "bad" is semantically similar to "poor" (Mikolov et al., 2013). These relationships are especially useful when trying to group similar customer feedback together as well as to understand differences in language between customers.

PART 4 — Information Extraction

- Natural Language Processing (NLP) techniques including named entity recognition (NER) enable the conversion of unstructured customer feedback to organized structured information. Within a customer product feedback application, the goal is to find the entities of interest in an unstructured source of information and organize the information in a way that provides insight into business decisions. Among all of the commonly applied NLP technologies for named entity recognition (NER) is one of the most popular. It enables identification of entities that have been identified with a name and classification as either person, place, date, time, organization, etc. (Nadeau and Sekine, 2007).
- A typical use case for this technology in a product/brand-based application would include finding entities associated with the name of a product or brand reference(s) along with potential location-based references or delivery references. A review that states "This Samsung phone has an excellent battery but was delivered too late to Malta" includes a product name ("Samsung phone") and a location ("Malta"). As a result, identifying these types of entities allow organizations to determine which product's they are discussing and what problems exist by geography or supplier.
- The implementation of a NER technology typically involves the use of a pre-trained NER model (such as models found in many NLP libraries like spaCy) to identify common entity types. Due to the nature of customer feedback containing informal language, further refinement may require some form of rule-based filtering to increase accuracy.

- Validation of the extracted information is also very important. The quality of NER results can be determined through metrics that quantify how well the system identifies correct entities, such as precision, recall, and F1-score (Jurafsky and Martin, 2023).
- As far as time and location-based extraction, in this example, location-based extraction is more relevant than time-based extraction because it identifies where customers' have had issues with delivery or where there were regional comments made about their experiences.

PART 5 — Language Modelling Application

- Language modeling is a major area of study in NLP concerned primarily with developing statistical methods to predict sequences of words and understanding generalization patterns within the text. The use of language models will allow analysis of common pattern recognition among customers' product review comments and also enable generation of representative sample text which represents customers' opinions about the product (Jurafsky and Martin, 2023).
- The most basic form of language modeling is through the use of n-gram models. Probability calculations for each word depend upon the preceding $n - 1$ words. A bigram model ($n = 2$), for example, would analyze the relationship between "good" and "battery", creating expressions like "good battery". Trigrams ($n = 3$) would also be able to analyze additional relationships and expressions that include more context, such as "very good battery." Through their ability to recognize common patterns in customer feedback and determine how customers typically refer to products, these systems will be able to provide the framework necessary for identifying trends and patterns in customer feedback.
- Language model evaluation includes metrics, including entropy and perplexity, to assess its quality. Entropy is a measure of uncertainty when attempting to guess what the next word will be in a given sequence; the lower the value of entropy, the less uncertain you are when guessing the next word in a sequence. Lower values of entropy indicate that a model has fewer uncertainties in predicting the next word in a sequence. Perplexity is another metric based on entropy that provides information regarding how effectively a model predicts data it has never seen before. When perplexity is low, the model is able to

predict unseen data. As stated above, both of these metrics are critical in determining if a model is capable of recognizing patterns found within customer product review comments (Manning et al., 2008).

PART 6 — Parsing & Grammar Analysis

- Parsing and Grammatical Analysis are critical aspects of NLP that provide an understanding of the structure and meaning of text. Parsing provides the ability to understand the relationship between words in a sentence; thus, it enables the system to obtain additional meaningful data from customer review comments beyond the basic word frequency of a review (Jurafsky and Martin, 2023).
- Part-of-Speech (POS) tagging is one of the main methods that may be used to assign grammatical labels (such as nouns, verbs, etc.) to each word in order to analyze the structure of language. The use of POS tags allows for an examination of descriptive patterns in customer feedback. Adjectives (such as "Excellent", "Poor", "Slow") will most likely relate directly to a specific product feature and can act as an indicator of overall customer sentiment.
- For this application, Shallow parsing would be acceptable as the purpose of the system is primarily to determine if there are meaningful patterns present versus whether there is deeper syntactical information. Dependency parsing can help determine the relationships between words. For example, when analyzing the sentence "the battery life is very good", dependency parsing helps establish a link between "very good" and "battery life"; therefore, establishing that the product has a strong point (strength).
- The system utilizes a key methodology to generate adjective-noun pairs via POS tagging. Thus, the system can identify a pattern that exists when two or more words appear together. Examples include "Good Quality", "Bad Packaging", "Slow Delivery". These pairs can then be analyzed collectively to create groupings based upon similar attributes/characteristics among other reviews; thereby, creating a structure to examine both product strengths and product weaknesses in comparison to a method of merely examining keywords.
- WordNet can be utilized to improve the semantic knowledge base of the system through identification of synonyms and associations between words.(Miller, 1995).

PART 7 — Evaluation & Implementation Plan

- The necessity for evaluating customer product feedback systems exists so as to determine if the insights being generated are both reliable and beneficial for making decisions. Customer product feedback systems require multiple evaluations across various components of the natural language processing (NLP) pipeline, such as preprocessing, information extraction and language modeling.
- Evaluation of named entity recognition (NER) type information extraction performance can utilize typical metrics, i.e., precision, recall and F1-score, in order to measure the accuracy at which entities identified through the customer review process (Jurafsky and Martin, 2023) have been correctly identified. A higher level of precision indicates that each of the extracted entities was correctly identified; likewise, a higher level of recall indicates that most of the entities relevant to the customer review were captured. The evaluation of vector-based analysis can also include comparisons of the output from TF-IDF regarding feature importance as well as assessments of the quality of semantic similarity based upon cosine similarity.
- Regarding language modeling performance, two common evaluation metrics utilized to assess whether or not a given sequence of words was predicted accurately by the model are perplexity and entropy. If a lower perplexity value results, it is likely indicative of a model having a greater degree of understanding relative to sequential word patterns present in customer reviews (Manning, Raghavan and Schutze, 2008).
- Implementation-wise, the customer feedback system can be created utilizing Python along with popular NLP libraries such as NLTK, spaCy and/or scikit-learn. Following an implementation structure that incorporates a pipelined approach may provide flexibility in future enhancements. One potential enhancement would be to integrate advanced deep learning architectures such as transformer networks into the existing architecture for improved contextual understanding.
- Therefore, while the proposed customer feedback system provides a baseline methodology for analyzing customer feedback, continuous refinement and evaluation of the system's performance and scalability should occur.

References

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